

Problem Set 8

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1 Probability Theory

Problem 1: In class, we claimed that if X and Y are independent, then $E(XY) = E(X) * E(Y)$. Prove it. We'll start you off:

$$E(XY) = \sum_{r \in \{r_1 r_2 \mid r_1 \in \text{im}(X), r_2 \in \text{im}(Y)\}} P(XY = r) * r = \sum \sum \dots$$

Problem 2: The following is the likelihood that a given dish will show up at a given American Thanksgiving dinner, along with its fairly arbitrary deliciousness rating R .

Food	Frequency	Rating
Turkey	90%	6
Mashed Potatoes	80%	4
Stuffing	70%	1
Cornbread	50%	6
Green Bean Casserole	60%	1
Sweet Potato Casserole	60%	8
Pumpkin Pie	80%	8
Pecan Pie	40%	20

What is the expected total deliciousness rating of a thanksgiving meal?

Problem 3: Assume that the likelihood of each dish appearing is independent, except for Pecan and Pumpkin Pie, which only have a 25% chance of appearing at the same meal. What is the variance of a meal's deliciousness rating?

Problem 4: The human genome contains 22 chromosome pairs, excluding the ones that determine their sex. For each pair, one is randomly drawn from the mother's two corresponding chromosomes and one from the father's two chromosomes (i.e. two options for each). Prove that the likelihood that two siblings have fewer than 16 chromosomes in common is less than one third. (You might observe that it's actually less than one sixth.)